

IBADAN LIFE: Deep Research & Pre-Production Bible

Executive Summary

The Sub-Saharan African mobile gaming market is undergoing a major structural shift. The viral emergence of web-based localized simulations—most notably *Lagos Life*, which accrued over three million registered accounts and peak concurrent user (CCU) counts exceeding 100,000 within days of its launch—demonstrates an immense, untapped demand for culturally authentic, socially driven virtual worlds. However, first-generation browser titles suffer from severe architectural bottlenecks, including vulnerability to client-side state exploits, text-centric user interfaces, static gameplay loops, and non-spatial social interactions.

This document establishes the pre-production blueprint and technical architecture for **IBADAN LIFE**, a 3D mobile-first multiplayer life simulation built on Unreal Engine 5. Set in a compressed, stylized representation of Ibadan, Nigeria—a metropolis defined by its iconic rust-tinted roofs, vibrant academic corridors, and sprawling commercial hubs—*IBADAN LIFE* transitions the regional simulation genre from a 2D web-based clicker into a spatial online ecosystem.

Unlike traditional open-world titles that rely on non-player character (NPC) populations to simulate vitality, *IBADAN LIFE* operates on a **Real-Player-First Design Principle**. Real human players form the core economic, structural, and social backbone of the world. NPCs are relegated strictly to infrastructural roles such as baseline municipal services, high-level civil law enforcement, and background ambient traffic.

By combining the physical agency of *Grand Theft Auto*, the emergent biological and lifestyle management of *The Sims*, the empathetic, proximity-based social discovery of *Sky: Children of the Light*, and the user-driven economic interdependence of *Roblox*, *IBADAN LIFE* introduces a novel sub-genre: the **Interdependent Urban Simulation**.

This report provides an exhaustive dissection of reference titles, field research on Ibadan's geographic and cultural matrix, detailed system specifications for life simulation and career supply chains, an authoritatively validated multiplayer network architecture, UE5 rendering pipelines optimized for mid-to-low-tier mobile hardware, and an automated AI-assisted asset production workflow.

Project Definition & Design Philosophy

Vision Statement

IBADAN LIFE is a persistent, 3D mobile-first multiplayer social life simulator designed to reflect the real social, economic, and operational dynamics of contemporary urban Nigeria. The title enables players to construct distinct identities, establish interdependent commercial enterprises, navigate authentic career trajectories, acquire real estate, and foster organic social networks within a shared spatial world.

The Real-Player-First Axiom

Traditional open-world games rely heavily on artificial intelligence to populate their environments, using hundreds of ambient NPCs to create an illusion of activity. In *IBADAN LIFE*, this paradigm is inverted. The server infrastructure, zone sharding, and interaction design prioritize human player density above all else.

NPCs do not simulate civilian life; they provide foundational framework services that cannot reliably depend on human availability. If a player dines at a local *Amala* joint, the owner, the cook, the logistics driver delivering yams, and the surrounding patrons are real players. System NPCs handle only high-level tasks, such as municipal traffic enforcement or emergency medical transport when no player medic is active.

Core Design Pillars

1. **Authentic Cultural Identity:** The game world eschews generic African urban tropes in favor of hyper-specific Ibadan visual markers, Yorùbá linguistics, regional culinary traditions, local transport networks, and distinct social etiquettes.
2. **Economic Interdependence:** No player operates in an economic vacuum. Upstream resource production, midstream refining/processing, and downstream retail services are bound together through player-to-player trade contracts and labor markets.
3. **Low-Friction Proximity Socialization:** Social discovery is seamless, spatial, and organic, utilizing proximity voice chat, expressive skeletal emotes, and collaborative activities designed to foster long-term digital relationships.
4. **Mobile Performance Parity:** High visual fidelity is maintained on mid-tier mobile hardware through aggressive architectural optimizations, modular asset streaming, and lightweight shader execution paths.

Lagos Life Deep Dive

To engineer a superior 3D title, an exhaustive analytical breakdown of the existing market baseline—*Lagos Life*—is required. Developed by Shalom Rayhamen (operating as Eliy) and launched in October 2026, *Lagos Life* demonstrated remarkable product-market fit while simultaneously revealing severe structural vulnerabilities.

System Architecture Summary

- **Platform:** Browser-Based Progressive Web App (PWA) built on HTML5, JavaScript, and WebSockets.
- **Population Metrics:** Over 3 million registered accounts; peak concurrent users exceeding 109,000.
- **Core Mechanic:** Menu-driven text interfaces tracking needs and monetary increments.
- **Monetization:** "Bachs" payment gateway converting real fiat into non-withdrawable virtual currency ("Game Naira").
- **Native Advertising:** Real-world brand promotion via in-game virtual billboards and rental plots.

Systemic Feature Breakdown

Feature Category	Documented Functionality in <i>Lagos Life</i>	Evidence Level
Character Creation	Basic 2D avatar selection; random financial starting profiles categorized as "Nepo" (privileged background with starting capital) or "Lapo" (microfinance debt/struggling baseline).	[VERIFIED]
Life Simulation	Six core needs tracked via HUD bars: Hunger, Energy, Hygiene, Bladder, Fun, and Social. Working degrades energy and hunger; neglected needs reduce overall productivity.	[VERIFIED]
Careers & Skills	Menu-selected jobs (e.g., freelancing, remote office work, local services) that yield virtual currency upon timer completion. Skills gate progression to higher salary tiers.	[VERIFIED]
Economy & Sinks	Fictional "Game Naira" earned through work or bought with real fiat via "Bachs" payment gateway. Main sinks include rent, food, entertainment, and luxury items.	[VERIFIED]
Housing	Static property renting; recurring lease payments due periodically; basic UI customization.	[VERIFIED]
Transportation	Abstract transit choices (Danfo, Okada, walking) represented via travel timer delays or instant menu transitions.	[STRONG EVIDENCE]
Multiplayer / Social	Browser chat rooms, direct messaging, peer-to-peer virtual money transfers (frequently manifesting as in-game begging).	[VERIFIED]
Advertising Engine	Native commercial monetization via virtual billboard rentals and branded virtual plots sold to real-world businesses.	[VERIFIED]

Feature Category	Documented Functionality in <i>Lagos Life</i>	Evidence Level
Security & Tech	Web browser access (PWA architecture), device fingerprinting, soft account deletion, non-encrypted messaging. Exploits previously allowed massive unauthorized currency generation.	[VERIFIED]

Architectural Inferences

From public statements and technical behavior, *Lagos Life* relies on a centralized Node.js/Python web server architecture paired with a relational database (e.g., PostgreSQL) communicating over WebSockets or HTTP REST APIs. The application state was historically evaluated heavily client-side, which enabled early currency duplication exploits until server-side validation was backfilled. Graphics rendering relies on lightweight 2D canvas elements, enabling broad compatibility across low-spec smartphones without dedicated GPU overhead.

Strategic Evaluation: Keep, Adapt, Improve, Reject

1. Keep / Adapt:

- *Socio-Economic Profiles*: The "Nepo" vs. "Lapo" starting conditions resonate deeply with regional audiences. Adapt this into a robust character background system with distinct stat modifiers, starting debts, and family networks.
- *Peer-to-Peer Cash Transfers*: Direct transfer mechanics facilitate social friction, extortion, generosity, and micro-commerce. Adapt with spatial wallet interactions (e.g., physically handing over physical cash or using an in-game mobile banking app).
- *In-World Native Advertising*: Real-world business integrations through in-game billboards provide sustainable revenue alongside player subscriptions.

2. Improve:

- *Spatial Environment*: Upgrade from static 2D web interfaces to a dynamic, fully explorable 3D city powered by Unreal Engine 5.
- *Career Mechanics*: Replace timer-based text clicks with active mini-games, physical transport, manual craft labor, and corporate management.
- *Social Interactions*: Transition from generic IRC-style web chat rooms to dynamic spatial proximity voice chat, contextual body language, and expressive avatar interaction.

3. Reject:

- *Client-Side State Validation*: Reject any client-authoritative state logic to prevent currency duplication and speed hacking.
- *Abstract Transport Timers*: Eliminate instant menu-based travel; transportation must involve physical movement across the 3D map via walking, public buses (*Danfo*), light taxis (*Micra*), or private vehicles.

Comparative Benchmark of Reference Titles

To contextualize *IBADAN LIFE* within the international gaming industry, four benchmark titles were analyzed across technical, mechanical, and social vectors.

Comparison Matrix

Game Title	Technical Architecture	Core Mechanical Strength	Primary Weakness	Strategic Takeaway for IBADAN LIFE
GTA V / Online	Proprietary RAGE Engine; Peer-to-Peer / Dedicated Hybrid	Spatial physical agency, vehicle physics, open-world streaming	High toxicity, severe server-client authority vulnerabilities, massive memory footprint	Adopt spatial open-world layout and vehicle handling; avoid P2P networking in favor of dedicated authoritative servers.
The Sims 4	Proprietary Single-Player C++ Engine	Interdependent needs engine, lifestyle routines, building grids	Completely non-spatial, single-player isolation, static instanced lots	Adapt contextual needs decay (Hunger, Energy, Social) to drive daily routines; avoid isolated lot loading screens.
Sky: Children of the Light	Custom Mobile Engine; Authoritative Cloud Instancing	Proximity-based social discovery, low-toxicity communication, mobile optimization	Minimal economic depth, abstract narrative, limited customization	Utilize spatial instancing and expressive non-verbal gestures to encourage friction-free player encounters.
Roblox	Proprietary Lua Engine; Cloud Micro-Services	Modular avatar ecosystem, user-generated micro-economies	Visual inconsistency, fragmented retention, vulnerable social engineering	Implement modular skeletal avatar assembly and user-operated commercial enterprises within a unified art style.

Comparative Analysis

Grand Theft Auto V / GTA Online

Grand Theft Auto V provides the gold standard for spatial urban agency, allowing players to navigate seamlessly between foot movement, vehicular transport, and interior spaces. However, GTA Online relies on a hybrid peer-to-peer networking model that suffers from severe security flaws and latency issues on volatile mobile networks. *IBADAN LIFE* adopts GTA's physical city

navigation but enforces strict server-authoritative state management.

The Sims Series

The Sims demonstrates the power of biological and social needs (Hunger, Energy, Hygiene, Fun) in establishing meaningful daily gameplay rhythms. When needs decay, players are forced to seek restoration. In *IBADAN LIFE*, needs are integrated directly into the multiplayer economy: an hungry player cannot simply click an infinite fridge; they must purchase food from a restaurant operated by another player.

Sky: Children of the Light

Sky proves that mobile hardware can support highly emotional, high-density social interactions when networking is engineered around spatial proximity. By placing players in shared spatial instances and providing seamless interaction prompts (such as lighting candles or sitting on benches), Sky removes social anxiety. *IBADAN LIFE* incorporates Sky's proximity discovery rules to make meeting strangers in Ibadan feel natural and safe.

Roblox

Roblox succeeds by turning players into creators and entrepreneurs, giving them tools to generate wealth within virtual experiences. *IBADAN LIFE* applies this economic model to urban businesses: players do not merely work for static system jobs; they can lease real estate, recruit human employees, manage inventories, and build commercial empires.

Field Research: Ibadan Context & Real-World Synthesis

IBADAN LIFE must not be a generic virtual city adorned with Nigerian names. It requires absolute geographic, structural, and social fidelity to the real-world city of Ibadan, Oyo State.

Geography & Urban Structure

Ibadan is defined by its sprawling topography, constructed across seven major hills, dominated by a vast expanse of rust-brown corrugated iron roofs interspersed with modern commercial developments and lush tropical vegetation.

Key Districts & Gameplay Functions

1. **Bodija:** Upscale residential zone and commercial hub. Contains high-end housing, lounges, boutique stores, and the bustling Bodija Market (agricultural wholesale trade).
2. **Dugbe:** The historic Central Business District (CBD). Features high-rise corporate office structures, banking hubs, the Dugbe Train Station, and major retail centers.
3. **University of Ibadan (UI) Corridor & Samonda:** The academic and youth engine. Characterized by high student density, affordable housing, computer/tech markets, fast-food outlets, and energetic nightlife.
4. **Iwo Road & Challenge:** Massive transport interchanges. Dominated by motor parks,

inter-state bus terminals, roadside vendors, informal mechanical workshops, and dense traffic gridlock.

5. **Agodi & Secretariat:** Administrative core containing the Oyo State Government Secretariat, Agodi Gardens (recreational park), and judicial complexes.
6. **Bower's Tower / Historic City Center (Oke-Are, Oje, Sabo):** The heritage core situated on the city's highest elevation point. High residential density, traditional Yoruba compound layouts, rust roofs, and local craft markets (e.g., Sabo watermelon/cattle trade).

Architecture & Environmental Visual Identifiers

- **Residential:** Traditional mud-brick plastered structures with weathered rust-red corrugated zinc roofs alongside modern gated compounds featuring electric fences, overhead water tanks (Geepee tanks), and diesel generator houses.
- **Commercial:** Roadside wooden kiosks (*Kiosk/Store*), container shops, multi-story concrete commercial plazas, open-air canopy markets, and modern glass-and-steel bank branches.
- **Infrastructure:** Overhead electricity poles with chaotic, tangled wiring arrays (*NEPA/IBEDC lines*), drainage gutters lining unpaved secondary roads, and yellow-and-blue painted concrete bus stops.

Cultural, Linguistic, and Social Dynamics

- **Linguistics:** English mixed with Yorùbá and Ibadan-specific Pidgin. Key slang integrated into system dialogue includes:
 - *"Alagba"* (Respectful term for an elder or respected individual).
 - *"Sere"* (To play/fool around).
 - *"Oga"* (Boss/Leader).
 - *"Pako"* (Fresh/Wealthy person, contrasted with *"Mopo"*).
 - *"Sharp-sharp"* (Expediently).
- **Culinary Culture:** Food acts as a primary life-simulation needs recovery mechanic. Key establishments include local *Bukas* serving *Amala Abula* (black yam flour with *Ewedu* and *Gbegiri* bean soup), *Asun* (spiced roasted goat meat), and modern fast-food chains.
- **Social Ceremonies:** Weekend *Owambe* parties featuring traditional live music, lavish native wear (*Aso-ebi*), and money-spraying mechanics that act as social wealth distribution events.

Transit Fleet Specifications

- **Nissan Micra Taxis:** Yellow-and-blue painted compact cars serving as the primary intra-city transit. Low cost, high density, erratic driving behavior.
- **Danfo (Volkswagen Transporter Buses):** Yellow commercial buses servicing major arterial routes.
- **Okada (Motorcycles) & Keke (Tricycles):** High-maneuverability vehicles navigating narrow, unpaved secondary residential alleys.

Game World Topology & Spatial Design

To achieve mobile performance without sacrificing environmental immersion, the real geographical layout of Ibadan (spanning over 3,000 square kilometers) undergoes a **Spatial Compression Ratio (SCR)** scaling of approximately 50:1.

Spatial District Layout

The game world is organized around a compressed central axis:

- **North Sector:** UI Student Hub & Samonda Tech Corridor (high youth density, low-cost housing, academic hubs).
- **Central Core:** Agodi Green Zone & Secretariat (government offices, public parks, central administrative facilities).
- **West Sector:** Dugbe CBD (corporate high-rises, banking plazas, rail infrastructure).
- **East Sector:** Bodija District (upscale estates, luxury dining, high-end nightclubs).
- **South Sector:** Iwo Road Interchange (transportation terminals, motor parks, mechanic workshops).
- **South-West Elevation:** Bower's Historic Hill (heritage compounds, rust-brown roof expanses, winding hillside roads).

Playable District Technical Specifications

District Name	In-Game Area (\text{km}^2)	Primary Land Use	Dynamic Player Density Target	Key Architectural Landmarks
Bodija Central	0.8	Upscale Residential / Commercial	64 Players / Instance	Bodija Market, Luxury Nightclubs, Gated Estates
Dugbe CBD	0.6	Commercial / Corporate / Finance	48 Players / Instance	Cocoa House (Compressed), Dugbe Mall, Banking Plaza
UI Student Hub	0.7	Academic / High-Density Low-Cost	96 Players / Instance	UI Main Gate, Student Hostels, Tech Kiosks
Iwo Transit Hub	0.5	Transportation / Wholesale Trade	64 Players / Instance	Interstate Bus Park, Mechanic Village, Roadside Market
Bower's Historic	0.4	Low-Income / Heritage Residential	32 Players / Instance	Bower's Tower Monument, Rust Roof Neighborhoods
Agodi Green Zone	0.3	Administrative / Recreational	32 Players / Instance	Secretariat Complex, Agodi Lake Park, Golf Club

Modular Player Avatar Architecture

Representing thousands of distinct human players on mobile GPUs requires an optimized

Modular Mesh Assembly System. Rather than rendering unique, fully rigged high-poly characters for every player, characters are dynamically composited from shared skeletal bases and texture atlases.

Avatar Technical Specifications

- **Rigging:** Single humanoid skeletal master rig containing 62 bones (avoiding per-finger bones on lower LODs to conserve skeletal evaluation time on mobile CPUs).
- **Polycount Budgets:**
 - **LOD 0 (Inventory Inspection):** 15,000 Triangles.
 - **LOD 1 (Proximity < 10m):** 6,000 Triangles.
 - **LOD 2 (Mid Distance 10m–30m):** 2,500 Triangles.
 - **LOD 3 (Far Distance > 30m / Mass Crowds):** 800 Triangles (Static Mesh impostor / simplified rig).

Modular Customization Pipeline

Avatars utilize a **Runtime Mesh Merging (FSkeletalMeshMerge)** architecture. When a player customizes their character, the server merges selected slot meshes (Head, Torso, Legs, Shoes) into a single skeletal mesh component on the client, dropping character draw calls from 8–10 down to **1 single draw call per character**.

Customization Categories

1. **Morph Targets:** Facial customization governed by 16 scalar parameters (jaw width, nose height, melanin slider, lip fullness, cheekbone prominence).
2. **Traditional Attire:** *Agbada* (flowing robes), *Iro and Buba*, *Gele* (headties), and *Fila* (caps) using custom *Aso-Oke* and *Ankara* fabric shaders.
3. **Contemporary Apparel:** Urban streetwear, university varsity jackets, corporate suits, mechanic overalls, and transport union uniforms.

Life Simulation Systems & Needs Engine

To create compelling daily routines without mimicking the rigid micro-management of *The Sims*, *IBADAN LIFE* utilizes a contextual, non-fatal **Dynamic Needs System**.

Mathematical Decay & Impact Model

Needs decay continuously over real-time minutes according to linear differential formulations:

$$N_t = N_{t-1} - (\alpha \cdot \Delta t)$$

Where N_t is the need level ($0 \leq N_t \leq 100$), α is the specific decay coefficient per minute, and Δt is elapsed time in minutes.

Need Variables & System Dependencies

Need Variable	Decay Rate (α per min)	Depletion Penalty ($N_t = 0$)	Recovery Mechanisms
Hunger	0.83 (100% loss over 2 hrs)	Max Movement Speed reduced by 40%; Stamina regeneration halted.	Consuming food items (<i>Amala, Jollof, Meat Pie</i>).
Energy	0.55 (100% loss over 3 hrs)	Character passes out for 60s; work earnings reduced by 50%.	Sleeping in owned/rented bed; drinking coffee/energy drinks.
Hygiene	0.41 (100% loss over 4 hrs)	Visible odor particle effect; locked out of entering luxury venues (Bodija lounges).	Showering at home; utilizing public restrooms.
Social	0.27 (100% loss over 6 hrs)	Experiencing "Loneliness" debuff: -15% XP/Skill progression rate.	Interacting with players (Voice, Chat, Emotes, Co-op work).

Economic Integration

Needs never directly result in character permadeath. Instead, they act as **economic and efficiency gates**. A player who maintains high need thresholds (>80%) receives a **"Soft Life" Buff**, granting +15% movement speed, +20% work wage output, and enhanced crafting luck. Neglected needs force players to spend money at player-owned eateries, barbershops, and relaxation spots, accelerating monetary velocity between players.

Interdependent Career & Economic Systems

The core gameplay thesis rejects menu-driven single-player job timers. All careers are integrated into a **Multi-Tier Interdependent Supply Chain**.

Multi-Tier Supply Chain Flow

1. **Tier 1: Raw Production:** Agricultural players harvest yams and cassava at rural farms on the city outskirts.
2. **Tier 2: Logistics & Haulage:** Freight drivers load raw crops into haulage trucks and transport them to central markets (Bodija Market).
3. **Tier 3: Enterprise Processing:** Restaurant owners purchase raw yams at market rates and cook them into prepared meals (*Amala Abula*).
4. **Tier 4: Retail & Service:** Hired player waitstaff serve cooked meals to patrons inside the restaurant.
5. **Tier 5: End Consumption:** Real player customers purchase and consume meals to satisfy their Hunger need and obtain performance buffs.

Career Taxonomy

Informal & Entry-Level Careers

- **Okada / Keke Rider:** Players mount motorcycles or tricycles, accept transport requests via an in-game smartphone app ("*RideNaija*"), and physically transport passengers to navigation markers.
- **Roadside Snack Vendor:** Players set up temporary physical stalls (*Suya* stand, roasted corn) and complete cooking mini-games balancing heat and timing.

Mid-Level Formal Employment

- **Bank Teller / Loan Officer:** Operates at Dugbe financial plazas, managing player credit applications, verifying micro-loans, and processing cash deposits.
- **LASTMA / Traffic Enforcement Officer:** Manages intersections, issues fines for illegal parking, and clears traffic jams to optimize delivery travel times.

Professional & Entrepreneurial Careers

- **Logistics Operator:** Owns haulage trucks or delivery vans, contracting with farms to supply urban markets.
- **Eatery / Lounge Proprietor:** Leases real estate, customizes kitchen layouts, sets menu prices, hires player staff, and manages commercial profit margins.

Player-Owned Businesses & Supply Chain Dynamics

Businesses are physical locations occupying real estate within the 3D map, requiring active management, inventory maintenance, and human employment.

Business Taxonomy & Economic Requirements

Business Type	Capital Requirement (Game Naira $\text{GN}\mathbb{N}$)	Primary Revenue Model	Required Supply Inputs	Supported Player Roles
Local Buka / Restaurant	$\mathbb{N}150,000$	Food retail sales to hungry players	Raw Produce (Yams, Meat, Oil, Vegetables)	Owner, Chef, Waiter, Supply Driver
Auto Mechanic Workshop	$\mathbb{N}300,000$	Vehicle repair, engine tuning, repainting	Spare Parts, Tools, Oil Drums	Workshop Manager, Chief Mechanic, Apprentice
Bodija Nightclub / Lounge	$\mathbb{N}1,200,000$	Door tickets, drink markups, VIP booth rents	Alcohol/Beverage Crates, DJ Equipment	Club Owner, DJ (Custom Audio Stream), Bartender, Security
Logistics Firm	$\mathbb{N}750,000$	Cargo transport	Delivery Vans,	Fleet Director,

Business Type	Capital Requirement (Game Naira $\text{\text{GN}}\mathbb{N}$)	Primary Revenue Model	Required Supply Inputs	Supported Player Roles
	0	contracts	Diesel Fuel, Storage Warehouses	Haulage Drivers

Property, Real Estate, and Housing Infrastructure

Real estate in *IBADAN LIFE* operates under a **Hybrid Instance-Physical Model** to balance open-world spatial presence with technical server memory limits.

Housing Architectural Model

- **External World (Persistent Map):** Physical building facades, entrance doors, address plates, outdoor balconies, and perimeter walls exist persistently in the shared open world.
- **Single-Family Properties:** Non-instanced, seamless physical transitions when opening exterior doors.
- **Multi-Unit Apartments:** Dynamic seamless instancing via portal volumes when stepping through corridor doors.

Housing Hierarchy

1. **Student Room (Samonda/UI):** Low-cost single-room apartment ("Face-Me-I-Face-You" layout) with shared courtyard facilities.
2. **Standard Flat (Challenge/Mokola):** Two-bedroom apartment unit with private bathroom, kitchen access, and customizable furniture grid.
3. **Luxury Gated Villa (Bodija Estates):** Non-instanced high-end compound featuring a private generator house, swimming pool, perimeter walls, and multi-car garage.

Vehicle & Mobility Framework

Vehicles provide both personal mobility and economic utility. Handling physics are powered by UE5's **Chaos Vehicles Plugin**, tuned for mobile arcade controls.

Vehicle Specs & Physics Tuning

Vehicle Type	Max Speed ($\text{\text{km/h}}$)	Acceleration Profile	Suspension Damping	Primary Operational Role
Nissan Micra	110	Moderate	Soft (Bouncy)	Intra-city Taxi Transport
Danfo Bus	90	Low	Heavy	High-Capacity Mass Transit
Bajaj Keke	60	High (Short Burst)	Very Soft	Short-Distance Alley Transit

Vehicle Type	Max Speed (\text{km/h})	Acceleration Profile	Suspension Damping	Primary Operational Role
Okada Motorcycle	100	High	Stiff	Fast Express Delivery
Luxury SUV	160	High	Balanced	Private Personal Transit

Maintenance & Wear Mechanics

Vehicles experience mechanical degradation over time:

- **Fuel Depletion:** Gas consumption scales with distance traveled. Gas stations across the map are player-owned or municipal stops.
- **Engine & Tire Wear:** Collisions and rough terrain degrade engine health and tire pressure. Low tire pressure impairs steering responsiveness, while engine health below 20% causes random stalling, requiring a player Mechanic to repair.

Social Fabric & Proximity Interaction Systems

Social interaction draws key lessons from *Sky: Children of the Light*, prioritizing spatial awareness, organic discovery, and expressive visual cues.

Spatial Interaction Mechanics

- **Proximity Voice Chat:** Powered by Vivox / EOS Voice. Voice audio attenuates linearly over a 15-meter spherical radius, with sound occluded by static building geometry.
- **3D Spatial Text:** Speech bubbles render directly above player avatars in 3D space, avoiding HUD clutter.
- **In-Game Smartphone ("IbadanConnect"):** A unified mobile phone UI featuring direct messaging, bank transfers, gig listings, and local event news feeds.

Culturally Rooted Social Systems

- **Aso-Ebi Group Buff:** When three or more players attend an event wearing matching group clothing patterns, they receive a **"Community Synergy" Buff**, increasing resistance to Fun and Social decay by 50%.
- **Money Spraying Mechanic:** At parties or street jams, players can execute a "Spray Money" action, emitting physical low-value cash notes into 3D space. Nearby players who collect these notes gain instant cash and fun restoration, driving organic social gatherings.

Server Architecture & Network Synchronization

To ensure stability across mobile network conditions (3G/4G/5G transitions), the server architecture employs an **Authoritative Dedicated Server Model with Spatial Sharding**.

Network Topology Architecture

- **Server Infrastructure:** Headless UE5 Dedicated Servers hosted on Linux container clusters managed via Kubernetes and Agones orchestration.

- **Replication Subsystem: UE5 Iris Replication System.** Iris delivers parallelized state serialization, dirty-object filtering, and reduced bandwidth consumption compared to legacy reflection replication.
- **Synchronization Frequencies:**
 - *Player & Vehicle Movement:* 20 Hz (client-side prediction with server reconciliation).
 - *Needs & Life Simulation:* 0.2 Hz (updated once every 5 seconds).
 - *Financial & Inventory Transactions:* HTTPS REST API calls with transactional database ACID compliance.

Scalability Scenarios

Deployment Scale	Local Instance Target	District Shard Target	Total Server Architecture
100 Players	64 Players	100 CCU	1 Dedicated Server Instance
1,000 Players	64 Players	250 CCU (4 Shards)	10 Dedicated Server Instances
10,000 Players	64 Players	2,500 CCU (40 Shards)	Kubernetes Cluster Pool
100,000 Players	64 Players	25,000 CCU (400 Shards)	Multi-Region Cloud Deployment

Unreal Engine 5 Systems Architecture

The technical implementation leverages specific subsystems natively available within Unreal Engine 5 (UE5.4–5.8 releases).

Subsystem Evaluation & Selection

Subsystem	Selection Status	Technical Justification
Iris Replication	SELECTED	Parallelizes actor replication, significantly reducing server CPU thread bottlenecks during high player density.
Mass Runtime	SELECTED	Simulates ambient pedestrian crowds and road traffic using Data-Oriented ECS with minimal CPU overhead.
Gameplay Ability System (GAS)	SELECTED	Handles needs attributes, status buffs, debuffs, and career skill trees with built-in networking prediction.
Chaos Vehicles	SELECTED	Provides stable, customizable vehicle physics for cars, buses, and motorcycles on mobile devices.
Enhanced Input	SELECTED	Maps touch-screen gesture

Subsystem	Selection Status	Technical Justification
		inputs dynamically while supporting physical gamepad hardware.
MetaSounds Engine	SELECTED	Generates dynamic procedural environment audio and spatial soundscapes efficiently.
World Partition	SELECTED	Streams static open-world geometry tiles asynchronously as players move across districts.

Mobile Optimization Strategy & Target Performance Budgets

Mobile engineering requires strict hardware targeting across Android (Adreno/Mali GPUs) and iOS devices.

Hardware Performance Targets

Parameter	Low-Tier Target (Entry Android)	High-Tier Target (Flagship Devices)
Target System RAM	4 GB RAM	8 GB+ RAM
Target Frame Rate	30 FPS Stable	60 FPS Stable
Max Draw Calls / Frame	< 1,200 Draw Calls	< 2,500 Draw Calls
Max Frame Triangles	< 1.5 Million Triangles	< 4.0 Million Triangles
Texture Compression	ASTC 8x8 Compression	ASTC 4x4 Compression
Rendering Pipeline	Mobile Forward+ Renderer	Mobile Forward+ Renderer
Shadow System	Static Lightmaps + CSM (20m Capped)	Mobile Dynamic CSG Shadows

Mobile Rendering Pipeline Decisions

- **Lumen / Nanite Policy: DISABLED FOR MOBILE TARGETS.** Global illumination relies on pre-baked lightmaps generated via GPU Lightmass for static geometry, combined with dynamic Cascaded Shadow Maps (CSM) capped at 20 meters. Nanite is replaced by standard static mesh LODs with aggressive distance culling.
- **Anti-Aliasing:** Subpixel Morphological Anti-Aliasing (SMAA) or Temporal Super Resolution (TSR Mobile) enabled via r.Mobile.AntiAliasing=5 to reduce specular aliasing on corrugated metal roofs.

AI-Driven Asset Generation & Production Pipeline

To streamline 3D asset generation and avoid manual modeling bottlenecks in Blender, an **Automated AI Asset Pipeline** is deployed.

Production Pipeline Sequence

1. **Source Capture:** High-resolution photography of real-world Ibadan street props (e.g., roadside kiosks, plastic chairs, local signage).
2. **AI Mesh Generation:** Images are processed via Meshy API or Tripo3D API to generate base 3D meshes and PBR texture maps in under 60 seconds.
3. **Automated Optimization:** Headless Python scripts in Blender automatically retopologize meshes to target polycounts (LOD 0–3) and unwraps UV maps.
4. **Skeletal Rigging:** Humanoid character assets pass through Meshy AI Auto-Rigging or AccuRIG APIs to generate 62-bone mobile skeletons with weighted skinning.
5. **UE5 Ingestion:** Automated Python scripts import assets into Unreal Engine, assign ASTC texture compression, generate material instances, and build colliders.

Toolchain & Licensing

Generative tools (Meshy/Tripo3D) are accessed via enterprise subscription tiers, granting royalty-free commercial usage rights for all generated meshes and textures.

Visual Art Direction & Environmental Aesthetics

The visual direction follows **Stylized Photorealism**—combining realistic architectural proportions and lighting with simplified, high-contrast texture palettes optimized for mobile screens.

Environmental Color Palette

- **Rust-Brown (#8B4513):** Dominates the roofs of older residential districts and exposed clay soil patches.
- **Dust-Orange (#E08D49):** Atmospheric height fog during Harmattan haze conditions and warm afternoon lighting.
- **Tropical Green (#2E8B57):** Lush foliage, roadside plantain trees, and courtyard vegetation.
- **Concrete Gray (#808080):** Asphalt roadways, concrete drain gutters, and cement block walls.
- **High-Saturation Accents (#FF1493 / #00FFFF):** Minibus paint stripes, commercial storefront signs, and market umbrellas.

Shading & Atmospheric Effects

- **Harmattan Season Simulation:** Adjusts atmospheric height fog levels dynamically, casting an orange dust haze across distant hills during afternoon hours.
- **Vertical Dust Shader:** Master materials include a global Z-axis world normal mask that automatically overlays red clay dust onto building bases and vehicle tires.

Audio Design & Proximity Soundscapes

Audio design uses spatialized ambient layers to reinforce real-world city immersion.

Audio Systems Layering

1. **Background Ambience:** Procedural MetaSounds graphs playing continuous distant cicada calls, generator hums, and tropical wind.
2. **Urban Density Layer:** Dynamic audio triggers emitting minibus horns, market haggling chatter, and street speaker announcements.
3. **Spatial Voice Layer:** Vivox proximity voice streams attenuated linearly based on distance and occluded by building walls.
4. **Broadcast Layer:** Local Afrobeat and Fuji music tracks emitting from speakers at player-owned Bukas, lounges, and vehicles.

UI/UX Framework for Mobile Interfaces

UI layouts prioritize mobile ergonomics, placing primary movement and action controls within natural thumb reach zones.

HUD Screen Layout Architecture

- **Top-Left Corner:** Needs Status HUD (Hunger, Energy, Hygiene bars) and cash balance display.
- **Top-Right Corner:** Mini-map display and toggle button for the IbadanConnect smartphone interface.
- **Bottom-Left Thumb Zone:** Virtual analog joystick for character movement and sprint locking.
- **Bottom-Right Thumb Zone:** Contextual Interaction Button, Emote Radial Wheel, and Push-to-Talk Proximity Mic toggle.

Contextual Action Button

A single dynamic interaction button adapts based on what the player is targeting:

- **Target = Vehicle Door:** Displays "Enter Vehicle".
- **Target = Restaurant Counter:** Displays "Order Food".
- **Target = Real Player:** Displays "Interact / Trade".

Player Safety, Moderation, and Governance

Maintaining a safe online environment requires robust moderation systems designed to enforce rules and meet legal requirements.

Governance & Security Protocols

1. **Age Verification:** Mandatory birthdate confirmation during account registration to enforce the **18+ Age Requirement**, complying with *Lagos Life* standards and social sandbox regulations.
2. **Data Privacy Compliance:** Adherence to the **Nigeria Data Protection Act (NDPA)** and

international privacy standards. User chat logs and device fingerprint hashes are encrypted using AES-256. Non-encrypted text channels are explicitly noted to users in the Terms of Service.

- 3. **Automated Moderation:** On-device NLP filters analyze local text chat for toxicity across English, Pidgin, and Yorùbá profanity lists. Voice streams are monitored via automated speech-to-text toxicity engines.
- 4. **Social Reputation & Penalties:** Players who engage in persistent toxicity or unprovoked disruption incur reductions in their **Social Reputation Score**. Players with low reputation scores are restricted from trading and may be assigned community service tasks (such as cleaning public areas in Agodi Park) before regaining full economic privileges.

Economy Engineering & Ethical Monetization

To maintain economic stability, *IBADAN LIFE* balances monetary inflows (faucets) and monetary outflows (sinks).

Macro-Economic Flow Architecture

- **Currency Faucets:** Wages paid for entry-level work (delivery, transportation), municipal relief stipends for new players ("Lapo" relief funds), and enterprise earnings from system sales.
- **Currency Sinks:** Housing lease payments, utility bills (electricity and generator diesel), food and hygiene items, vehicle maintenance and fuel, and a 2.5% municipal transaction tax on peer-to-peer wallet transfers.

Ethical Monetization Framework

Pay-to-win mechanics are strictly prohibited. Monetization focuses entirely on non-pay-to-win revenue streams:

Monetization Category	Implementation Model	Economic Impact
Cosmetics & Fashion	Purchases for native outfits (Aso-Oke weaves), vehicle paint wraps, and hairstyles.	Purely visual customization; zero gameplay stat boosts.
Furniture Season Pass	Seasonal interior decor themes (Modern Minimalist, Traditional Royal Court).	Unlocks decorative blueprints; stat recovery rates match base furniture equivalents.
In-Game Billboard Ad Network	Real-world businesses buy native billboard space through an ad portal.	Generates real-world corporate revenue without imposing costs on players.
Business Storefront Customization	Custom neon signage and unique building facades for player shops.	Increases shop visibility without altering product sale prices or stats.

Comprehensive Gameplay Loop Hierarchy

Player retention is driven by nested feedback loops operating across different timescales.

Loop Hierarchy Mechanics

- **30-Second Loop (Moment-to-Moment):** Spatial navigation, avoiding traffic, scanning nearby players, observing proximity chat bubbles.
- **5-Minute Loop (Immediate Needs):** Purchasing food at a nearby Buka to satisfy Hunger, taking a short rest, completing a single transport delivery run.
- **30-Minute Loop (Session Goals):** Completing a career work shift, earning wages, repairing a vehicle, meeting friends at a lounge.
- **Daily Loop (Routine Maintenance):** Paying housing rent and utility bills, restocking business inventory, attending weekend social events (*Owambes*).
- **Weekly Loop (Economic Expansion):** Upgrading real estate leases, buying additional delivery vehicles, expanding business enterprises.
- **Long-Term Loop (Identity & Dominance):** Acquiring prime property in Bodija, building a business empire, establishing a top social reputation.

MVP Scope & Phased Feature Roadmap

To prevent scope creep, development is divided into four distinct phases.

Feature Roadmap Matrix

Feature Domain	MVP 1 (Proof of Concept)	MVP 2 (Expansion)	Post-MVP	Full Vision
Playable Map Area	UI Student Hub & Samonda Corridor	Add Dugbe CBD & Iwo Road	Add Bodija Estate & Agodi Park	Full Ibadan Metro Area (Bower's, Airport)
Server Density	32 Players / Instance	64 Players / Instance	128 Players / Instance	Dynamic Spatial Sharding Across City
Avatar System	Modular Base Rig + 20 Outfits	Expanded Apparel + Native Wear	Custom Morph Sliders + Hair Physics	Full Character Accessories & Tattoos
Careers	Okada Driver, Buka Staff, Vendor	Mechanics, Bank Tellers, Traffic Police	Logistics Operator, Real Estate Agent	Corporate Executive, Event Promoter
Vehicles	Bajaj Keke, Micra Taxi	Danfo Buses, Cargo Vans	Luxury Sedans, Sports Cars	Heavy Semi-Trucks, Helicopters
Core Systems	Basic Needs, Local Chat, Housing Rent	Full Supply Chain, Player Shops	District Governance Elections	Native Billboard Ad Web Network

Technical Development Roadmap

Development follows a strict dependency order to ensure backend systems support higher-level features.

Phase 1: Core Engine & Network Infrastructure

- Dedicated Server Setup (UE5 + Linux/Kubernetes)
- Iris Replication Pipeline Setup
- User Authentication & Database Persistence Layer

Phase 2: Avatar & Movement Systems

- Modular Humanoid Rig Assembly & FSkeletalMeshMerge Implementation
- Character Customization Interface
- Enhanced Touch Input Controls & Camera Mechanics

Phase[span_40] (start_span) [span_40] (end_span) 3: Environment & World Streaming

- World Partition Tile Setup
- UI Student Hub District Architecture Modeling
- Pre-baked GPU Lightmass Lighting Setup

Phase 4: Gameplay & Life Simulation

- Gameplay Ability System (GAS) Needs Attributes
- Item Crafting, Storage, & Inventory Systems
- Chaos Vehicles Setup (Micra, Keke)

Phase 5: Interdependent Economy & Social Mechanics

- Property Leasing & Business Storefront Logic
- Multi-tier Supply Chain Contracting Systems
- Proximity Voice Chat Integration (Vivox / EOS)

AI Developer Workflow Integration

To maximize developer efficiency, AI tools are integrated into technical workflows:

- **Code Generation:** GitHub Copilot and Claude assist in drafting UE5 C++ subsystems, GAS attributes, and Python Blender automation scripts.
- **Automated QA Testing:** CI/CD build scripts deploy headless automated bot clients on dedicated server instances, stress-testing network replication, memory usage, and zone transitions under peak load.

Answers to Critical Pre-Production Questions

1. **Product Differentiation:** *IBADAN LIFE* moves away from 2D browser menus to deliver a persistent 3D open world built on Unreal Engine 5, combining real-time spatial navigation with rich regional identity.
2. **Daily Player Activity:** Players manage real-time life needs, work interdependent jobs, run physical businesses, transport cargo, customize properties, and socialize via proximity voice chat.
3. **Real-Player Population:** High player density is prioritized through server sharding and proximity discovery rules, while system NPCs handle only essential infrastructure, law enforcement, and background traffic.
4. **Economic Interdependence:** Supply chains link players together: raw crop producers

- rely on freight drivers, who supply restaurant owners, who employ waitstaff to serve real player customers.
5. **Multiplayer Architecture:** Headless UE5 Dedicated Servers on Kubernetes run the Iris Replication System, using dynamic spatial sharding to keep local instance loads stable at 64 players.
 6. **Mobile Optimization:** High performance is achieved by disabling Lumen and Nanite, using pre-baked GPU Lightmass lighting, merging character meshes at runtime, and applying ASTC texture compression.
 7. **Visual Quality:** Using Stylized Photorealism alongside pre-baked static lighting allows the game to deliver atmospheric visual quality on mobile devices.
 8. **MVP Scope:** The initial MVP focuses on the UI Student Hub and Samonda Corridor, supporting 32 concurrent players per instance with core transport and food supply chain systems.
 9. **Efficient Avatar Production:** Dynamic runtime mesh merging (FSkeletalMeshMerge) combines character clothing components into a single skeletal mesh, reducing draw calls to 1 per avatar.
 10. **Exploit-Resistant Economy:** All business management, inventory counts, and monetary transactions are validated authoritatively on the server via ACID-compliant REST APIs, preventing currency duplication.
 11. **Social Retention:** Proximity voice chat, group synergy buffs (*Aso-Ebi* bonuses), and physical money-spraying interactions encourage organic player gatherings and long-term social connections.
 12. **AI Workflow Automation:** AI generative pipelines (Meshy/TripAPIs) convert real-world photos into optimized, auto-rigged 3D assets, streamlining world-building without manual Blender modeling.
 13. **Cost Efficiency:** Leveraging open-source container tools, cloud services, and AI asset pipelines minimizes initial capital requirements during early pre-production.
 14. **Scalability:** The backend scales smoothly from single dedicated server instances up to multi-region cloud container clusters using Kubernetes and Agones orchestration.

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